Noman Saffat Sajid

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Education

North South University September 2021 - September 2025
Bachelor of Science in Computer Science and Engineering (BSCSE) ***CGPA:*** *3.87 out of 4.00*

Research Experience

Developed a preprocessing module to reduce deep learning model cost through cropping 2024 - Ongoing
Under the supervision of Dr. Md. Shahriar Karim, Assoc. Prof. ECE, NSU

- Developed a preprocessing module that reduces training time and computational expense (CO_2 emission, and energy consumption) by 30-50% across different deep learning architectures, with only a 1-3% decrease in accuracy.
- Developed a Principal Component Analysis based statistical algorithm for image cropping dimension selection.
- Contributed substantially to paper writing, figure creation, and experimental analysis of the proposed method.

Mathematical model for open set recognition September 2025 - May 2026
Under the supervision of Dr. Mohammad Shifat-E-Rabbi, Asst. Prof. ECE, NSU

- Performed benchmarking of proposed method against existing deep learning and machine learning methods, and demonstrated competitive or superior performance of the proposed method.
- Led the paper writing, figure creation, and experimental analysis of the proposed method.

Publications

N. S. Sajid, M. R. Khan and M. S. Karim, 'Objects Cannot Hide Behind The Shades – We Can See Through and Localize', *Under review at Transactions on Machine Learning Research (TMLR). (Presented as a non-archival paper at the European Conference on Computer Vision (ECCV) CDEL Workshop 2026)*

N. S. Sajid, M. Rahman and M. Subha, M. Shifat-E-Rabbi, S. Rahman and G. K. Rohde, 'A Sliced-Wasserstein Distance Based Approach for Open Set Recognition', *under review at Conference on Neural Information Processing Systems (NeurIPS) GDDL workshop 2026*

N. S. Sajid and M. S. Karim, 'Target Localization in Cell-Based Image Analysis and Disease Diagnosis', *Published in International Conference on Learning Representations (ICLR) LMRL Workshop 2025*

F.M A. Hossain, **N. S. Sajid** and R. M. Rahman, 'Semantic-Preserving Image Compression and Restoration Pipeline for Bandwidth-Constrained UAV Applications: A Task-Aware Evaluation Framework', *Published in Artificial Intelligence and Applications (AIA) 2026*

Course Projects

Image Processing Pipeline for Efficient Transmission of UAV Images | *Senior year project*

- Designed and implemented a pipeline that uses color quantization as a compression strategy for faster image transmission, which achieves 68% file size reduction and 3× faster transmission times.
- Implemented a deep learning model that restores the image after transmission, from quality degradation caused by color quantization.

IoT-Based Smart Home Security and Monitoring System | *Embedded Systems Course Project*

- Interfaced the STM32 blue pill microcontroller with bluetooth module, OLED display and keyboard.
- Implemented a mobile app using MIT app inventor, initialized a database, and interfaced with the Bluetooth module.

Work Experience

Research Assistant (RA) September 2025 - May 2026
Under Dr. Mohammad Shifat-E-Rabbi, Asst. Prof. ECE, NSU

- Worked on a mathematical model for open set recognition

Extracurricular Experience

DB Hackathon: Database Design Challenge October 2023
2nd prize winner *NSU, Dhaka*

- Solved different situation-based problems using Entity-Relationship diagrams.
- Wrote complex SQL queries using joins, subqueries, and aggregate functions, to solve competition problems.

Skills

Languages: English and Bengali
Programming Languages: Python (OpenCV, PyTorch, NumPy, Matplotlib), C and Java
Operating Systems: Linux and Windows